

An Exact Algebraic All-Mode Intertwining for $SU(2)$ Heat Transport

Machine-checked Haar spectral coordinates and the finite Dobrushin quotient

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Abstract

We give a machine-checked spectral-coordinate theorem for the concrete $SU(2)$ heat operator on normalized Haar probability. The coefficient object is the algebraic direct sum $\mathcal{A} = \mathbb{N} \rightarrow_0 \mathbb{R}$, so one statement covers every finite real character expansion without a fixed mode cutoff. Its lift $\Phi : \mathcal{A} \rightarrow \{f : SU(2) \rightarrow \mathbb{C}\}$ is injective; Haar pairing recovers every coefficient and gives the exact finite-support Euclidean pairing. For $t > 0$, actual convolution by the infinite heat kernel intertwines Φ with the diagonal Casimir step

$$(D_t a)_n = \exp\left(-\frac{tn(n+2)}{4}\right) a_n, \quad H_t \Phi = \Phi D_t.$$

This upgrades the preceding two-mode bridge to all algebraic character modes. The previously formalized two-state Dobrushin transfer remains an exact quotient of the $\{0, 1\}$ sector: its states contain genuine bundled elements of $SU(2)$, its finite lift is bijective, and its actual operator intertwines with $\text{diag}(1, e^{-3t/4})$. The infinite-kernel identity follows from finite character orthogonality and dominated convergence, not from an eigenvalue hypothesis. The result is an exact invariant algebraic sector plus a finite conjugacy on its first two modes, not a Peter–Weyl completeness theorem for completed $L^2(SU(2))$ and not a claim that $SU(2)$ is finite. Every load-bearing identity is checked in Lean 4 against pinned revisions.

Artifact statement. The new formal source is `YangMills/OS/SU2HeatAllModes.lean`. Its concrete dependencies are `YangMills/OS/SU2HeatIntertwining.lean`, `YangMills/OS/SU2HeatTransport.lean`, the pinned satellite and `YangMills/OS/SU2TransportWitness.lean`. Exact revisions, byte counts, line-ending hashes, build commands, and runtime information are recorded in the release manifest distributed with this paper.

1 Problem and contribution

For a compact gauge group, heat-kernel convolution is the natural transfer mechanism behind the exact solution of two-dimensional Yang–Mills theory. At the formal level, however, three distinctions matter.

First, an abstract transfer theorem is not yet an inhabited gauge-theoretic endpoint. Second, equality of one scalar decay rate is not an operator map. Third, a numerical certificate for a matrix identity is not a Lean witness inhabiting $SU(2)$. The purpose of the present construction is to close precisely this interface by a commuting diagram.

The contribution is the following chain, checked without a project-local axiom and without assuming any of its conclusions:

- (i) finite heat-kernel partial sums act diagonally on every character already present in the sum;

- (ii) dominated convergence upgrades this to the actual infinite heat kernel;
- (iii) the Casimir ordering $3 \leq n(n+2)$ for $n \geq 1$ identifies χ_1 as the sharp slowest non-vacuum mode;
- (iv) evaluation at the identity proves $\chi_1 \neq 0$ exactly;
- (v) all finite-support real character expansions are embedded through one exact Haar-isometric lift, and coefficient extraction proves injectivity;
- (vi) the actual infinite-kernel heat operator intertwines this lift with the diagonal Casimir step on $\mathbb{N} \rightarrow_0 \mathbb{R}$;
- (vii) the first two coefficient modes map bijectively to the full finite transfer space, with the fundamental coefficient mapping to the sign vector; and
- (viii) both operators commute with the same diagonal step, yielding an actual restriction and conjugacy theorem in Lean, while the earlier nonzero projected gap endpoint remains available as a corollary.

The novelty is deliberately narrower than the exact area-law results already available in the satellite. We do not rename those results as new. Instead, we add the missing spectral map needed by the parent transport endpoint.

2 Canonical endpoint map

Table 1 records the exact division of labor. This is also the minimum interpretation that connects the continuous heat kernel to the published parent theorem. A rate equality alone is too weak; a full completed $L^2(\text{SU}(2))$ spectral theorem is stronger than necessary. The exact middle object is the algebraic direct sum of all character modes, together with its two-character finite quotient. Completed- L^2 completeness is neither needed for the parent endpoint nor silently assumed.

Layer	Exact object	Role
Satellite	normalized Haar probability, Chebyshev characters, finite and infinite class heat kernels, character convolution orthogonality	published analytic input
New heat bridge	<code>heatKernel_character_eigen</code> , Casimir-rate ordering, nonzero fundamental mode	continuous spectral input
New all-mode bridge	$\mathbb{N} \rightarrow_0 \mathbb{R}$, coefficient extraction, exact Haar pairing and diagonal heat intertwining	all finite character expansions in one theorem
Two-mode quotient	bijective finite lift and exact finite commuting identity	genuine finite restriction/conjugacy map
Frozen parent	two bundled $\text{SU}(2)$ holonomies, exact two-state covariance, <code>exact_transport_witness</code>	finite non-vacuous endpoint
Abstract endpoint	abstract uniform-gap theorem	converts exact band decay into a projected transfer norm bound

Table 1: Canonical endpoint map. The all-mode row is the new theorem lane; the two-mode quotient is the minimum connection to the parent endpoint.

The parent endpoint is not merely existential at the carrier level. Its type is a sigma type whose elements contain a finite label, a bundled 2×2 complex matrix, a proof of unitarity, a proof of determinant one, and an equality with one of two explicitly constructed holonomies. Hence every carrier element inhabits $\text{SU}(2)$ in Lean.

3 The concrete heat operator

Let μ_H be normalized Haar probability on $\text{SU}(2)$. For functions f, h the satellite convolution convention is

$$(f * h)(g) = \int_{\text{SU}(2)} f(x)h(x^{-1}g) d\mu_H(x).$$

Write χ_n for the real-valued Chebyshev character of highest-weight label n , regarded as a complex function, and set

$$a_n(t) = \exp\left(-\frac{tn(n+2)}{4}\right).$$

The class heat kernel is the convergent series

$$K_t(g) = \sum_{n=0}^{\infty} (n+1)a_n(t)\chi_n(g).$$

The factor $n+1$ is the representation dimension, while $n(n+2)/4$ is the $\text{SU}(2)$ quadratic Casimir in the satellite's normalization.

The input orthogonality statement is already stronger than a scalar inner product identity. It gives the convolution formula

$$(\chi_n * \chi_m)(g) = \begin{cases} \chi_n(g)/(n+1), & n = m, \\ 0, & n \neq m. \end{cases} \quad (1)$$

This normalization cancels the dimension in K_t exactly.

3.1 Finite partial sums

Let $K_t^{<N} = \sum_{n < N} (n+1)a_n(t)\chi_n$. The first new Lean theorem is the finite identity below.

Theorem 3.1 (finite character eigenidentity). *For $m < N$, $t \in \mathbb{R}$, and $g \in \text{SU}(2)$,*

$$(K_t^{<N} * \chi_m)(g) = a_m(t)\chi_m(g).$$

Proof. Finite integrability follows from continuity on the compact group. Linearity of the Haar integral turns convolution with the partial sum into a finite sum of integrals. Equation (1) kills every term except $n = m$; the remaining factor $(m+1)$ cancels its reciprocal. The Lean proof expands the integrand before invoking the satellite theorem, so no desired eigenidentity is passed as a premise. $\square$

The formal statement is:

```
theorem heatKernelPartial_character_eigen_of_lt
  (N : Nat) (t : Real) (m : Nat) (hm : m < N) (g : SU2) :
  su2Convolution (su2HeatKernelPartial N t)
    (su2CharacterChebyshev m) g =
  su2ClassHeatWeight t m * su2CharacterChebyshev m g
```

3.2 Passage to the infinite kernel

The infinite theorem requires more than rewriting a formal series. For fixed m and g , define

$$F_N(x) = K_t^{<N}(x)\chi_m(x^{-1}g), \quad F(x) = K_t(x)\chi_m(x^{-1}g).$$

The satellite supplies pointwise convergence $K_t^{<N}(x) \rightarrow K_t(x)$ for $t > 0$ and a summable heat-kernel majorant. It also supplies $|\chi_m(y)| \leq m + 1$. Thus

$$|F_N(x)| \leq \left(\sum_{n=0}^{\infty} M_t(n) \right) (m + 1),$$

a constant integrable against the probability measure μ_H . Dominated convergence passes the limit through the Haar integral. For all $N \geq m + 1$, Theorem 3.1 makes that integral the same constant; uniqueness of limits completes the proof.

Theorem 3.2 (infinite heat-mode identity). *If $t > 0$, then for every $m \in \mathbb{N}$ and $g \in \text{SU}(2)$,*

$$(K_t * \chi_m)(g) = \exp\left(-\frac{tm(m+2)}{4}\right) \chi_m(g).$$

The crucial point is quantification over the actual bundled compact group and the actual infinite kernel. No finite carrier appears in Theorem 3.2.

4 Sharp fundamental mode

Define $r_n(t) = \exp(-tn(n+2)/4)$. The vacuum mode is $r_0(t) = 1$. For $n \geq 1$ the elementary inequality

$$3 \leq n(n+2)$$

and monotonicity of the real exponential give

$$r_n(t) \leq r_1(t) = e^{-3t/4} \quad (t \geq 0).$$

For $t > 0$, $0 < r_1(t) < 1$. Consequently the fundamental representation is the sharp slowest mode among all non-vacuum character labels. “Sharp” here means that equality is attained at $n = 1$, not that a Peter–Weyl completeness theorem for the completed class-function Hilbert space has been formalized.

Non-vacuity is exact. The satellite proves $\chi_n(1) = n + 1$; hence $\chi_1(1) = 2$, and function extensionality shows χ_1 cannot be the zero function. The specialization of Theorem 3.2 is therefore a genuine eigenmode:

$$(K_t * \chi_1)(g) = e^{-3t/4} \chi_1(g), \quad \chi_1 \neq 0. \quad (2)$$

5 The exact algebraic all-mode intertwining

Let $\mathcal{A} = \mathbb{N} \rightarrow_0 \mathbb{R}$ be the algebraic direct sum: an element a is a real sequence with finite support, but no global cutoff is fixed. Define

$$(\Phi a)(g) = \sum_{n \in \text{supp}(a)} a_n \chi_n(g),$$

$$(D_t a)_n = r_n(t) a_n, \quad r_n(t) = \exp\left(-\frac{tn(n+2)}{4}\right).$$

Since D_t cannot enlarge support, it is an endomorphism of $\mathcal{A}$. The point of using a single finitely supported object, instead of a theorem for each chosen N , is uniformity: every finite character sector and every finite linear combination is an instance of the same commuting diagram.

Theorem 5.1 (exact algebraic all-mode intertwining). *For every $t > 0$:*

(a) *coefficient extraction is exact,*

$$\int_{\text{SU}(2)} (\Phi a)(g) \chi_m(g) d\mu_{\text{H}}(g) = a_m \quad (m \in \mathbb{N});$$

(b) *Φ is injective and preserves the finite-support Haar pairing,*

$$\int_{\text{SU}(2)} (\Phi a)(g) (\Phi b)(g) d\mu_{\text{H}}(g) = \sum_{n \in \text{supp}(b)} a_n b_n;$$

(c) *actual normalized-Haar heat convolution obeys*

$$H_t \Phi = \Phi D_t.$$

No linear-independence or eigenvalue conclusion is passed as a hypothesis. Coefficient extraction expands a finite support sum, moves that sum through the Haar integral using proved integrability, and invokes the concrete orthogonality theorem. Injectivity follows by extracting every coefficient. For the commuting identity, the proof again moves only a finite sum through the integral, then uses Theorem 3.2 term by term. Thus the actual infinite heat kernel participates in the proof of every coordinate.

5.1 The exact two-mode finite quotient

Restrict $\mathcal{A}$ to coefficients supported on $\{0, 1\}$ and identify them with $E = \mathbb{R}^2$. The continuous lift becomes $a_0 \chi_0 + a_1 \chi_1$, and D_t becomes $\text{diag}(1, e^{-3t/4})$. The finite lift is

$$(\Phi_2 a)(x) = a_0 + a_1 \sigma(x), \quad \sigma(0) = 1, \quad \sigma(1) = -1.$$

The preceding formal module proves that Φ_2 is bijective onto the full two-state Euclidean transfer space and that the actual finite operator obeys $W_t \Phi_2 = \Phi_2 D_t$. Therefore the induced map from the $\{0, 1\}$ continuous sector to the finite space is a genuine operator conjugacy. This is the minimum restriction that connects Theorem 5.1 to the published parent transport endpoint.

Non-vacuity has two independent exact checks. The all-mode theorem recovers every coefficient by Haar integration. On the finite quotient, evaluation at the bundled phase $p = \text{diag}(i, -i)$ and at the identity recovers a_0 and a_1 , while $\Phi_2(0, 1)$ is the exact nonzero sign vector used in the projected-transfer witness.

6 Finite $\text{SU}(2)$ -inhabited transport witness

The finite endpoint is a quotient/interface object, not an approximation in which group axioms are discarded. Its two labels carry two explicit matrices: the identity and a noncentral diagonal phase with determinant one. Both are bundled as elements of $\text{SU}(2)$, and their inequality is proved exactly.

For a rate $0 < r < 1$, the transition kernel is

$$W_r = \frac{1}{2} \begin{pmatrix} 1+r & 1-r \\ 1-r & 1+r \end{pmatrix}.$$

Its rows sum to one, it is symmetric and strictly positive, and

$$W_r^k = W_{r^k}.$$

For any observable f on the two carrier points, the band covariance is the identity

$$\text{Cov}_k(f, f) = \left(\frac{f(0) - f(1)}{2} \right)^2 r^k. \quad (3)$$

Thus the decay premise consumed by the Dobrushin theorem is derived, not assumed. The sign observable $(1, -1)$ is orthogonal to the normalized constant vacuum, and the projected transfer sends its first component to r . Since $r > 0$, the projected operator is not zero.

The frozen witness instantiates r by

$$\text{heatRate}(t) = e^{-3t/4}.$$

The new bridge proves this is definitionally the same real number as $r_1(t)$ in the continuous heat theorem. It then combines Equation (2) with the existing endpoint:

Theorem 6.1 (continuous-to-finite transport bridge). *For $t > 0$:*

- (a) $K_t * \chi_1 = \text{heatRate}(t)\chi_1$ pointwise on $\text{SU}(2)$;
- (b) $\chi_1 \neq 0$;
- (c) there exists $m > 0$ for which the norm of the vacuum-projected finite transfer operator is at most e^{-m} ; and
- (d) that projected operator is nonzero.

Theorem 6.1 is retained because it is the published parent gap endpoint. Theorem 5.1 is stronger on the continuous side: it simultaneously identifies every finite character expansion with an exact diagonal coordinate vector. The finite matrix remains the quotient of the $\{0, 1\}$ sector only. It is not identified with the operator on completed $L^2(\text{SU}(2))$, and no two-state approximation of the compact group is claimed.

7 Formal proof architecture

The formal dependency graph is intentionally short:

```

satellite character convolution + heat majorant
      ↓
heatKernelPartial_character_eigen_of_lt
      ↓ dominated convergence
      heatKernel_character_eigen
      ↓ Casimir ordering +  $\chi_1(1) = 2$ 
      heatKernel_fundamental_eigen
      ↓ finite-support linearity + coefficient extraction
      exact_algebraic_all_mode_intertwining
      ↓ restriction to  $\{0, 1\}$  + two explicit lifts
heatOperator_continuousLift  finiteOperator_finiteLift
      ↓
      exact_two_mode_intertwining.

```

Table 2 performs the hypothesis-discharge audit requested for the endpoint.

The Lean proof uses classical choice and standard analytic facts inherited from Mathlib, as expected for Haar integration and infinite series. The audit criterion is therefore not an empty `#print axioms` list, but absence of `sorryAx` and absence of project-local axioms. The exact printed axiom sets are captured in the verification log.

Required fact	Source	Discharge
$t > 0$	bridge input	used for summability and $r < 1$
Haar integrability	compact continuity	satellite continuity lemmas and probability measure
dominating function	heat majorant	summable constant times $m + 1$
character orthogonality	satellite theorem	used termwise before the limit
Casimir ordering	new arithmetic lemma	proved by casting $1 \leq n$ and nonlinear arithmetic
$\chi_1 \neq 0$	new exact lemma	evaluation at the identity gives 2
continuous lift injective	new exact lemma	evaluation at the phase and identity; independently certified by Haar pairing
all-mode coefficient extraction	new exact lemma	finite-support integral linearity plus concrete Haar character orthogonality
all-mode invariance	new intertwining	termwise infinite-kernel eigenidentity on the support of each coefficient vector
continuous invariance	new intertwining	dominated integrability, integral linearity, and the two character eigenidentities
finite conjugacy	new intertwining	bijective coefficient lift and exact entrywise matrix algebra
finite kernel laws	frozen witness	explicit two-by-two algebra
decay premise	frozen witness	equality (3), not a hypothesis
projected nonzero	frozen witness	exact sign-vector evaluation

Table 2: Every load-bearing hypothesis and where it is discharged.

8 Reproducibility and frozen revisions

The artifact uses Lean 4 through the parent repository’s pinned toolchain. The satellite is fetched as an exact Lake dependency, not as an unversioned copy. At fabrication time the pins were:

Object	Revision
frozen predecessor	59c42d93ec5f4bf660792163c41407d9b1f3d8b4
lean-2d-yang-mills	05c4ec316cb9aa295416670a2578b1c2e77e1c36
Mathlib	07642720480157414db592fa85b626dafb71355b

The decisive commands are:

```
lake update
lake build YangMills.OS.SU2HeatTransport
lake build YangMills.OS.SU2HeatIntertwining
lake build YangMills.OS.SU2HeatAllModes
lake env lean Task48Axioms.lean
```

They were run in a fresh Google Colab Pro+ CPU/high-RAM runtime with no GPU. Local desktop work was limited to source editing, Git, hashing, TeX compilation and PDF rendering. A source file on disk was not counted as verified until its `.olean` had been materialized remotely.

9 Scope, limitations, and interpretation

The results support the following exact statement: the infinite Haar heat operator has an invariant algebraic sector containing every finite real character expansion; Haar pairing supplies exact spectral coordinates there; the heat operator is diagonal in those coordinates; the $\{0,1\}$ restriction is conjugate to the entire genuine $SU(2)$ -inhabited two-state transfer operator; and the same finite witness fires the parent projected-gap endpoint non-vacuously.

They do not establish any of the following:

- four-dimensional continuum Yang–Mills or the Clay mass-gap problem;
- a thermodynamic or continuum limit of the finite witness;
- Peter–Weyl completeness or density of the algebraic character span in the completed class-function Hilbert space;
- the spectrum on non-class matrix coefficients, or an operator-norm computation on the completed full Haar L^2 space;
- a new proof of the satellite’s Migdal recursion, semigroup law, or exact finite-cellulation area law; or
- that two carrier points exhaust, discretize, or approximate the compact group.

The right interpretation is an exact locally finite invariant restriction: each vector uses finitely many modes, but the formal coefficient object has no fixed dimension. It replaces a previously two-mode continuous endpoint by an injective all-mode lift and an exact diagonal intertwining. The finite parent still consumes only the first two coordinates. Hilbert completion and Peter–Weyl density remain separate work.

10 Relation to the literature

The use of heat kernels and character expansions in two-dimensional Yang–Mills goes back to Migdal’s recursion equations [1]. Driver constructed the Yang–Mills measure and related it to the heat equation on compact groups [2]; rigorous compact-surface developments include Sengupta [3] and Lévy [4]. The present paper does not reprove those broad analytic constructions. It contributes a small, exact, inspectable theorem interface inside Lean.

The formalization builds on Mathlib [5] and on the pinned two-dimensional satellite [6]. The repository citation is a commit permalink: it identifies the precise source used by Lake rather than a moving branch.

11 Conclusion

The continuous endpoint now covers every algebraic character mode in one Lean object. It is an actual Haar integral on bundled $SU(2)$; the infinite kernel is handled by dominated convergence; coefficient extraction proves non-vacuity; and the lift preserves the exact finite-support Haar pairing. The heat operator commutes with the full Casimir diagonal on this algebraic sector. The finite side still consists of distinct genuine holonomies; its two-coordinate lift is bijective, its covariance identity is exact, and its projected transfer is nonzero. Thus $e^{-3t/4}$ is the fundamental entry of a proved all-mode diagonalization and, on the first two modes, an exact Lean-checked finite conjugacy.

References

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- [2] B. K. Driver, “YM₂: continuum expectations, lattice convergence, and lassos,” *Communications in Mathematical Physics* 123 (1989), 575–616. doi:10.1007/BF01238812.
- [3] A. Sengupta, *Gauge Theory on Compact Surfaces*, Memoirs of the American Mathematical Society 126, no. 600 (1997). doi:10.1090/memo/0600.
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